A Lightweight Brain Tumor Segmentation Network Based on 3D Inverted Residual Modules

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ABSTRACT

Semantic segmentation technology based on deep learning has played an important role for doctors in identifying brain tumor regions and formulating treatment plans. Popular automated segmentation methods for brain tumors include 2D and 3D convolution networks. The 3D networks give better results but lead to a significant increase in parameters and computational cost. In this paper, we propose a lightweight brain tumor segmentation network composed of 3D inverted residual modules, which can significantly reduce the computational complexity of 3D models. Based on a lightweight depthwise separable convolution, our 3D inverted residual module extracts high-dimensional brain tumor features through an intermediate expansion layer, thus improving performance. On the brain tumor dataset BraTS 2018, our network achieves dice scores of 80.8%, 90.7%, and 84.3% (for ET, WT, and TC, respectively) with only 0.68M parameters and 51.46G FLOPs. The results show that our method can significantly reduce the complexity of the 3D model and achieve very competitive performance.

CCS CONCEPTS

• **Computing methodologies**; • **Artificial intelligence**; • **Computer vision**; • **Computer vision problems**; • **Image segmentation**;

KEYWORDS

Lightweight CNN, Brain Tumor Segmentation, 3D Inverted Residual Module

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1 INTRODUCTION

Glioma is the most common and highly harmful primary brain tumor. According to the development status of glioma, it is divided into high-grade glioma (HGG) and low-grade glioma (LGG). Low-grade gliomas are benign tumors that usually occur in younger patients. High-grade gliomas are highly malignant and dangerous. The task of brain tumor segmentation is to divide brain tumor tissue from normal brain tissue [2, 3]. Brain tumor image segmentation is critical for doctors to identify brain tumor regions and develop treatment plans. However, Manual segmentation by doctors is very time-consuming and laborious [1]. Automated brain tumor segmentation provides an efficient and precise solution for analyzing brain tumors.

In recent years, deep convolutional networks have demonstrated state-of-the-art performance on the task of brain tumor segmentation [2]. Ronneberger et al. [5] proposed U-Net, a convolutional network based on an encoder-decoder structure, which has made great strides in medical image segmentation. A series of improved methods based on 2D U-Net have been successfully applied to the task of brain tumor segmentation [1]. However, 2D networks pay little attention to the correlation of 3D slice sequences. 3D networks [6] can simultaneously encode 3D semantic information and achieve better results. However, the parameters and complexity of a 3D network increase several times, making it difficult to deploy on platforms with limited computing power. Currently, some works try to study lightweight networks [16–18] to solve the high complexity of 3D models. Although these lightweight networks could effectively reduce the computational complexity, they also drop the overall performance significantly. Consequently, designing a lightweight high-performance automatic segmentation model for brain tumors remains a challenging task.

In this work, we propose a lightweight brain tumor segmentation network based on 3D inversed residual modules. In our design, the

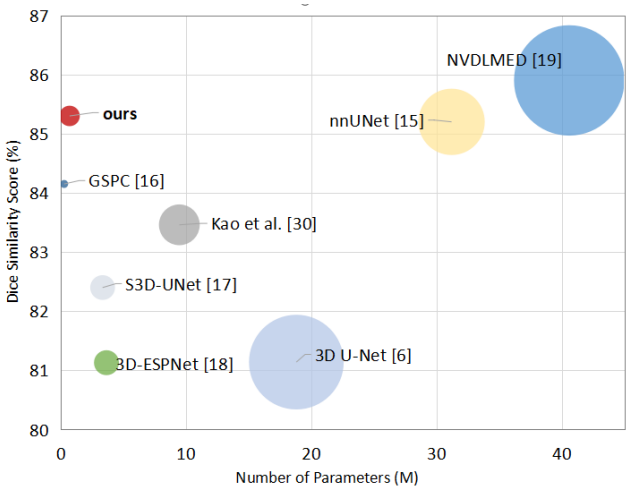


Figure 1: Performance comparison with other state-of-the-art models on BraTS2018 [34], where the vertical axis is the dice score, and the circle size indicates computational complexity (FLOPs). As shown in Figure 1, the dice score of our model is comparable to the high complexity models (nnUnet [15] and NVDLMED [19]) while still being lightweight.

3D inverted residual module can achieve competitive results by using a lightweight depthwise separable convolution to extract the feature information of high-dimensional brain tumors. Compared with state-of-the-art methods (Figure 1), our model achieves a good balance between accuracy and effectiveness.

2 RELATED WORK

Semantic segmentation network based on convolutions. The main objective of semantic segmentation is to classify images at the pixel level. Long et al. [4] deleted all fully connected layers in the classification network and defined a fully convolutional network (FCN), which was a milestone in semantic segmentation. FCN fuses features from shallow and deep layers of the encoder, which is beneficial for accurate segmentation. Ronneberger et al. [5] proposed U-Net (including the encoder, decoder, and bottleneck) based on FCN to achieve more refined segmentation. A larger image receptive field is more beneficial for the encoder to extract feature information. To enlarge the receptive field, many designs [20, 21] such as Atrous Convolution, Atrous Spatial Pyramid Pooling (ASPP), and Pyramid Pooling Module (PPM) have been proposed for tackling this issue. Besides, to avoid the local awareness of CNNs, some attention-based methods [22–24] have been proposed for obtaining more global contextual information.

Lightweight convolutions. In the early stages of deep learning, large convolution kernels were used to obtain larger receptive fields [25]. Later work found that using multiple small convolution kernels instead of large convolution kernels can expand the receptive field [26]. Small convolution kernels have fewer parameters than large ones and can use deeper networks under the same memory conditions [7, 26, 27]. Howard et al. [11] proposed a memory-efficient depthwise separable convolution based on depthwise and pointwise convolution [28], which significantly reduced computation and parameters. Studies are also dedicated to transforming convolutions, such as ShuffleNet [13, 14] or group shift pointwise convolution [16].

Brain Tumor Semantic Segmentation. BraTS is a large-scale brain tumor segmentation competition [2]. In recent years, CNN has performed better than traditional machine learning in the competition. Myronenko et al. [19] added a variational autoencoder to the U-Net to reconstruct the image and regularize the shared decoder. Jiang et al. [29] proposed a two-stage cascaded U-Net to obtain coarse-to-fine segmentation results. Isensee et al. [15] proposed a segmentation framework to fit the 3D medical image data, thus achieving competitive performance on many 3D medical segmentation tasks. However, 3D CNN requires a large computational resource overhead. To tackle this problem, some lightweight modules-based works have been proposed to reduce the high computational cost [16–18].

3 METHOD

The proposed network follows an encoding and decoding architecture (U-Net). Aiming at the problem of the high computational complexity of the 3D CNN network, We take the proposed 3D inverted residual module as the basic block of U-Net. The detailed design of the 3D inverted residual module is shown in Figure 2. In terms of the loss function, we adopt the dice loss and the cross-entropy loss to jointly optimize the model.

3.1 3D Inverted Residual Module

Depthwise separable convolution is a variant of standard convolution that can significantly reduce computational complexity [11]. It decomposes a standard convolution into a depthwise convolution and a pointwise convolution (As shown in Figure 2(a)). Each channel of the depthwise convolution operates with only one convolution kernel, which can significantly reduce the involved parameters. However, extracting each channel's feature information will inevitably destroy the corresponding information. In our design, the depthwise separable convolution can effectively utilize the information between channels by combining the depthwise and pointwise convolution.

The computational cost (FLOPs) of 2D standard convolution and 2D depthwise separable convolution is calculated below. Suppose the input feature map size is: $D_F \cdot D_F \cdot M$, and the output feature map size is: $D_F \cdot D_F \cdot N$, where M and N are the number of input and output feature map channels, respectively. The comparison of computational cost between standard convolution and depthwise separable convolution is:

$$\frac{D_K \cdot D_K \cdot M \cdot D_F \cdot D_F + M \cdot N \cdot D_F \cdot D_F}{D_K \cdot D_K \cdot M \cdot N \cdot D_F \cdot D_F} = \frac{1}{N} + \frac{1}{D_K^2} \quad (1)$$

When the convolution kernel size is 3 (K = 3), the computational cost of the depthwise separable convolution is only 1/9 to 1/8 of the standard convolution, which improves memory efficiency.

We propose a 3D lightweight module to replace the computationally expensive 3D standard convolution. The proposed 3D inverted residual module (As shown in Figure 2(b)) is based on the lightweight depthwise separable convolution. More feature information

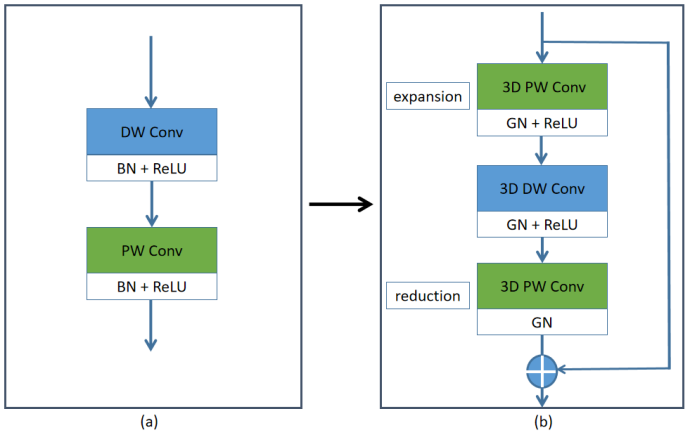


Figure 2: The proposed 3D inverted residual module (b), where 3D PW Conv indicates a 3D pointwise convolution, 3D DW Conv indicates a 3D depthwise convolution and GN indicates group normalization. The proposed module first inflates the dimension, extracts high-dimensional semantic features and then reduces the dimension, which is opposite of resnet [7].

is lost due to the use of nonlinear activation in low-dimensional space. The module first uses a 3D pointwise convolution to expand the feature information from low-dimensional space to high-dimensional space. The high-dimensional feature information is extracted by a lightweight 3D depthwise convolution and then projected back to the low-dimensional space. Finally, similar to the residual operation, output and input features are added to reduce network degradation. We also remove the ReLU layer after the second pointwise convolution to keep the linear mapping [12]. Group normalization [10] can solve the internal covariate shift problem better than batch normalization [9] when the batch size is small. Therefore, we use group normalization instead of batch normalization in our model.

3.2 Network Architecture

The overall design of our network (Figure 3) follows an encoder-decoder architecture based on 3D inverted residual modules.

Encoder. Similar to 3D U-Net, the encoder has three stages. After each stage, the channels of the feature map are doubled, and the size of the feature map is reduced by half. Each stage consists of two 3D inverted residual modules and a downsampling operation. The pooling operation can easily lose image boundary details. In the downsampling stage, we use a 3x3x3 standard convolution with a stride of 2 instead of a pooling operation. The size of the feature map image is changed from 4x160x192x128 to 128x20x24x16 after three downsampling stages.

Decoder. The decoder consists of three stages and performs the opposite of the encoder. After each stage, the channels of the feature map are halved and the size of the feature map is doubled. In the upsampling stage, we use a 1x1x1 pointwise convolution to halve the channels of the feature map, then use a trilinear interpolation function to double the feature map size, and finally add the output of the corresponding encoder stage. The size of the image feature map is changed from 128x20x24x16 to 3x160x192x128 after three downsampling stages. Finally, the network outputs a segmentation image with pixel class labels.

3.3 Loss Function

Our loss function includes the cross-entropy loss and the dice loss. Cross-entropy loss is the most commonly used loss function in classification tasks, and it drives the model to converge faster in the early training stages. For brain tumor images, the imbalance between the foreground and background samples is severe. The Tumor tissue is only a small part of the entire brain tissue. And there is also an imbalance between different types of tumor samples. Dice coefficient is a regional correlation function used to calculate the similarity of two samples, which can alleviate the problem of sample imbalance [8]. We combine the cross-entropy loss and the dice loss to optimize the parameters and achieve a better convergence effect.

$$L_{total} = L_{dice} + L_{bce} \tag{2}$$

$$L_{dice} = 1 - \frac{2 * \sum y_{gt} * p_{predict}}{\sum y_{gt}^2 + \sum p_{predict}^2 + \epsilon} \tag{3}$$

$$L_{bce} = \frac{1}{N} \sum - \left[y_{gt} * \log\left(p_{predict}\right) + (1 - y_{gt}) * \log\left(1 - p_{predict}\right) \right] \tag{4}$$

where y_{gt} is the true label of the pixel, $p_{predict}$ is the predicted probability of the pixel and ϵ is equal to 1e-5 to avoid zero division.

4 EXPERIMENTS

4.1 Experimental Details

We conduct experiments on the BraTS dataset (BraTs2018 and BraTs2019) [2, 34]. The BraTS 2018 dataset includes 285 training data (210 HGG and 75 LGG) and 66 validation data. The BraTS 2019 dataset includes 335 training data (259 HGG and 76 LGG) and 125 validation data. The training set contains four brain tumor modalities (FLAIR, T1, T1ce, and T2) and one label (Figure 4). The validation set only contains four MRI images. The validation results need to be submitted to the online website for verification. The size of each MRI image is 240x240x155. The regions for brain tumor segmentation included whole tumors (labels 1, 2, and 4), core tumors (labels 1 and 4), and enhancing tumors (label 1). The evaluation

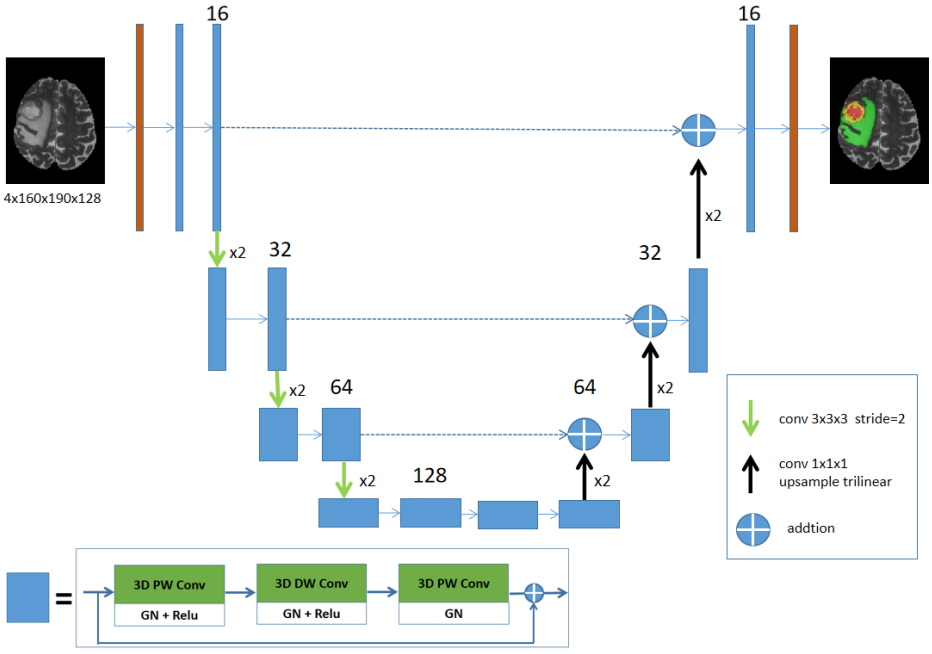


Figure 3: Overview of the network architecture.

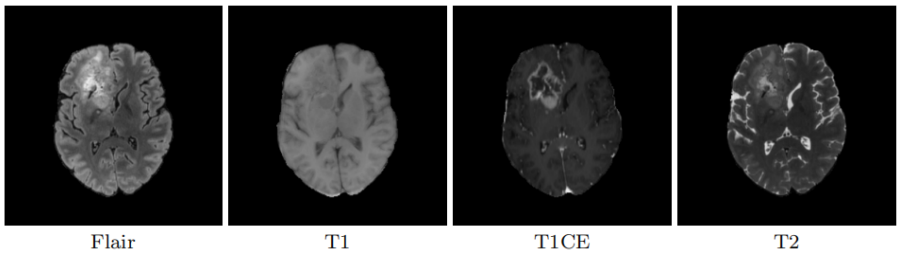


Figure 4: Visualization of brain tumor MRI images in 4 modalities

index of brain tumor segmentation is measured by dice score and Hausdorff95 distance.

Our experiments are performed on PyTorch and a single Tesla V100. The max-min normalization strategy is used to normalize the image pixel values between 0 to 1. We randomly crop each MRI data from 4x240x240x155 to 4x160x192x128 (concatenate four modalities as input) and randomly flip the three axes of the image with a probability of 0.5. We adopt the Adam optimizer strategy to train the model with one batch size for 300 epochs. To reduce overfitting, we use the L2 norm to regularize the parameters. After the encoder, neuron parameters are randomly dropped with a probability of 0.2 (Dropout). We set the initial learning rate to 1e-4 with poly learning rate decay.

$$\alpha = \alpha_0 * \left(1 - \frac{e}{N_e}\right)^{0.9} \tag{5}$$

where α_0 indicates the initial learning rate (1e-4), e indicates an epoch counter, and N_e indicates max epochs.

We test a full image rather than dividing patches. We believe that testing on the whole image is beneficial to improve segmentation results. To obtain efficient segmentation results, we only evaluate one model and do not apply test time augmentation (TTA). If there are multiply connected domains in the predicted image, we remove the connected domains less than a certain threshold of the maximum connected domain. And we replace enhancing tumors that are smaller than a certain threshold with necrotic tumors.

4.2 Comparison with Other Methods

In Table 1, we compare our method with other state-of-the-art brain tumor segmentation methods ([19, 30, 15, 6, 17, 18, 16]) on BraTS 2018 dataset. Compared to the 3D U-Net, our model achieves an average 2.5% increase in dice score and an average decrease of 1.41 in Hausdorff95 distance. Notably, we achieve the best dice score of WT with only 0.68M parameters and 51.46G FLOPs. Our method is 29× less computational complexity and 58× less parameters than the

Table 1: Performance on BraTS 2018 validation dataset.

Network	Params(M)	FLOPs(G)	Dice(%)			Hausdorff95(mm)		
			ET	WT	TC	ET	WT	TC
NVDLMED [19]	40.06	1495.53	**81.4**	90.4	**85.9**	3.80	4.48	8.27
Kao et al. [30]	9.45	203.96	78.7	90.4	81.3	3.81	**4.32**	7.56
nnUNet [15]	31.2	539.65	80.5	90.6	84.5	**2.53**	5.35	6.87
3D U-Net [6]	18.80	1099.48	77.9	90.1	80.3	3.08	6.89	9.17
S3D-UNet [17]	3.32	75.20	74.9	89.3	83.0	-	-	-
3D-ESPNet [18]	3.63	76.51	73.7	88.3	81.4	-	-	-
GSPC[16]	0.25	7.91	78.1	90.2	83.2	4.01	5.22	**6.53**
ours	0.68	51.46	80.8	**90.7**	84.3	2.69	4.48	7.73

Table 2: Performance on BraTS 2019 validation dataset.

Network	Dice(%)			Hausdorff95(mm)		
	ET	WT	TC	ET	WT	TC
3D U-Net [6]	70.8	87.3	72.4	5.06	9.43	8.71
V-Net [8]	73.8	88.7	76.5	6.13	6.25	8.70
KiU-Net [32]	73.2	87.6	73.9	6.32	8.94	9.89
Attention U-Net [31]	**75.9**	88.8	77.2	5.20	7.75	8.25
Wang et al. [33]	73.7	89.4	**80.7**	5.99	5.67	**7.35**
Ours	75.4	**90.3**	79.9	**3.68**	**4.81**	7.51

Table 3: Ablation experiment on BraTS 2018 validation dataset. "BL": 3D inverted residual module without residual connection and expansion ratio is 1; "RC": 3D inverted residual module with residual connection; "ER2": the expansion ratio of the 3D inverted residual module is 2; "ER4": the expansion ratio of the 3D inverted residual module is 4.

Network	Params(M)	FLOPS(G)	Dice(%)		
			ET	WT	TC
BL	0.49	35.67	75.1	89.0	78.3
BL+RC	0.49	35.67	75.4	89.9	78.5
BL+RC+ER2	0.68	51.46	**80.4**	90.0	**82.8**
BL+RC+ER4	1.06	83.05	78.2	**90.4**	82.3

state-of-the-art model NVDLMED [19], and our segmentation accuracy is very close to theirs. Among the lightweight methods(S3D-UNet [17], 3D-ESPNet [18], GSPC [16]), GSPC [16] has the fewest parameters and FLOPs, but their ET and TC dice scores drop significantly. Our method achieves a good balance between segmentation accuracy and model scale.

As shown in Table 2, compared with the methods of 3D U-Net [19], V-Net [8], KiU-Net [32], Attention U-Net [31], and Wang et al. [33], our method achieves excellent performance on BraTS 2019 dataset. As observed, we achieves the best Dice score for WT, and the Hausdorff95 distance for ET and WT are greatly reduced compared to other methods.

Figure 5 shows some examples of segmentation results on the BraTS 2018. We divide 80% of the training dataset for training and 20% for offline validation. For nnUNet, we train the model for 1000 epochs (following the default configuration of nnUNet). For 3D U-Net, it can not recognize the internal details of brain tumors. There are morphological domain gaps between HGG and LGG. And we find that nnUNet [15] performs well on HGG data but fails to capture the key feature information on LGG data. Our method performs well on HGG and LGG, showing robustness and effectiveness.

4.3 Ablation Experiment

We perform two ablation experiments on BraTS 2018 including studying the effectiveness of the 3D inverted residual module and comparing it with other lightweight modules. We train all models for 100 epochs.

We experiment with three expansion ratios to study the effectiveness of the 3D inverted residual module. The experimental results are shown in Table 3. When the expansion ratio of the 3D inverted residual module is 1, the performance of the model is poor. It can be seen that adding residual connections reduces the degradation of the network and improves performance. The performance is significantly improved when the expansion ratio is 2. It is shown that

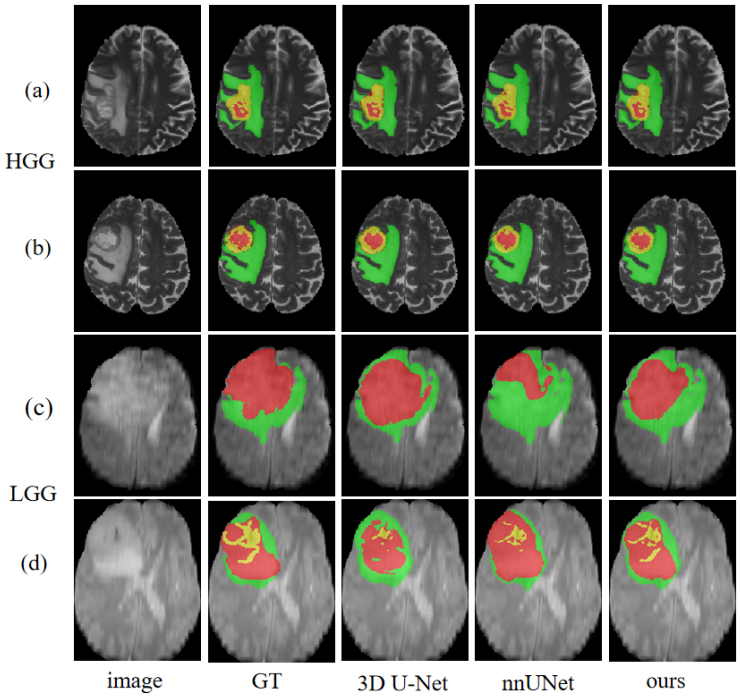


Figure 5: Visualization of segmentation results on BraTs2018 dataset, where (a) and (b) are high-grade gliomas (HGG), (c) and (d) are low-grade gliomas (LGG). The regions in green, red, and yellow indicate edema tumors, necrotic and non-enhancing tumors, and enhancing tumors, respectively.

Table 4: Comparison of the proposed method with other popular lightweight modules on BraTs 2018.

Network	Params(M)	FLOPs(G)	Dice(%)		
			ET	WT	TC
ShuffleNet [13]	1.27	70.20	80.2	89.8	81.6
ShuffleNet V2 [14]	1.40	79.83	79.0	89.6	82.7
MobileNet [11]	1.59	95.24	79.4	89.5	82.2
Ours	0.68	51.46	**80.4**	**90.0**	**82.8**

depthwise separable convolution extracts more feature information in high-dimensional space. When the expansion ratio is 4, the highest WT dice score was obtained, but the ET dice score dropped severely compared to the expansion ratio is 2. For a small medical dataset, too many parameters can easily lead to the overfitting of the model. We think an expansion ratio of 2 is the best solution.

We also replace the proposed 3D inverted residual module with other popular lightweight modules to verify the effectiveness. In Table 4, MobileNet [11] replaces standard convolution with pointwise convolution and depthwise convolution. ShuffleNet [13, 14] uses Channel Shuffling operation to increase the information flow between features. For fairness, the GPU memory for all experiments is limited to 12G. Table 4 shows our method achieves competitive performance with less parameters and computational cost.

5 CONCLUSION

In this work, we propose a lightweight 3D network for automated brain tumor segmentation, replacing the most commonly used convolution layers in the encoder-decoder architectures with 3D inverted residual modules. Notably, our method is 21×less computational complexity and 27× less parameters than 3D U-Net but achieves more remarkable performance. We also compare our method with state-of-the-art methods on BraTS dataset. The experimental results demonstrated that our design can achieve excellent efficiency and accuracy characteristics. In our future work, we plan to extend our method to other 3D medical segmentation tasks for broader research, further advancing the development of smart medicine.

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